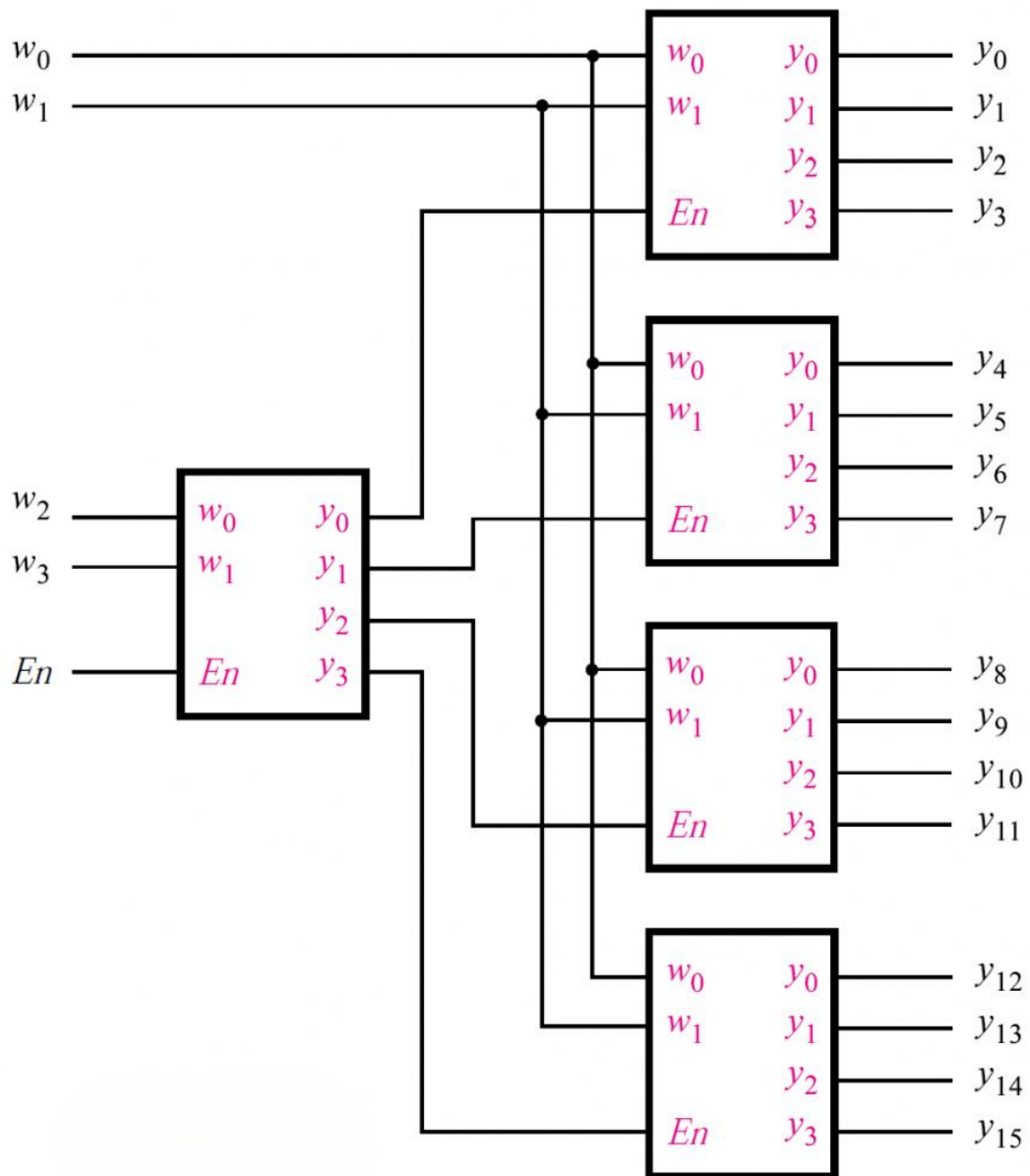


Day 15

4x16 Decoder using 2x4 Decoder in Verilog

Circuit Diagram :



Verilog Code

4:16 Decoder code using 2:4 Decoder [structural modeling]

```
1 module decoder2x4 (sel,Y,en);
2     input [1:0]sel;
3     input en;
4     output reg [3:0]Y;
5     always @(*) begin
6         if(en) begin
7             case (sel)
8                 2'b00: Y = 4'b0001;
9                 2'b01: Y = 4'b0010;
10                2'b10: Y = 4'b0100;
11                2'b11: Y = 4'b1000;
12            endcase
13        end
14        else Y=4'b0000;
15    end
16 endmodule

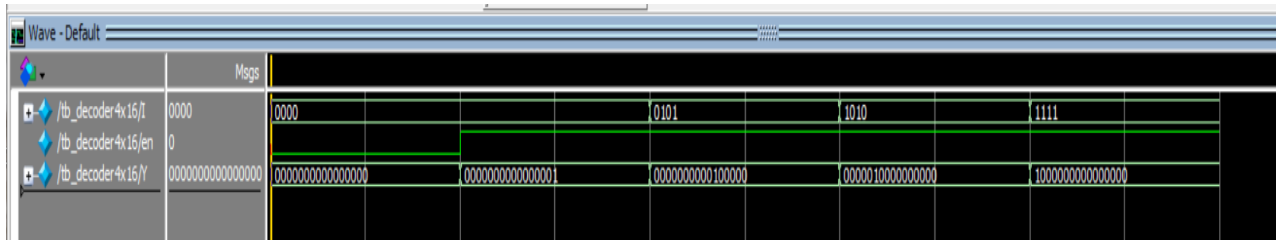
17
18 module decoder4x16(I,Y,en);
19     input en;
20     input [3:0]I;
21     output [15:0]Y;
22     wire [3:0]w;
23     decoder2x4 d0(I[3:2],w,en);
24     decoder2x4 d1(I[1:0],Y[3:0],w[0]);
25     decoder2x4 d2(I[1:0],Y[7:4],w[1]);
26     decoder2x4 d3(I[1:0],Y[11:8],w[2]);
27     decoder2x4 d4(I[1:0],Y[15:12],w[3]);
28 endmodule
```

Testbench Code

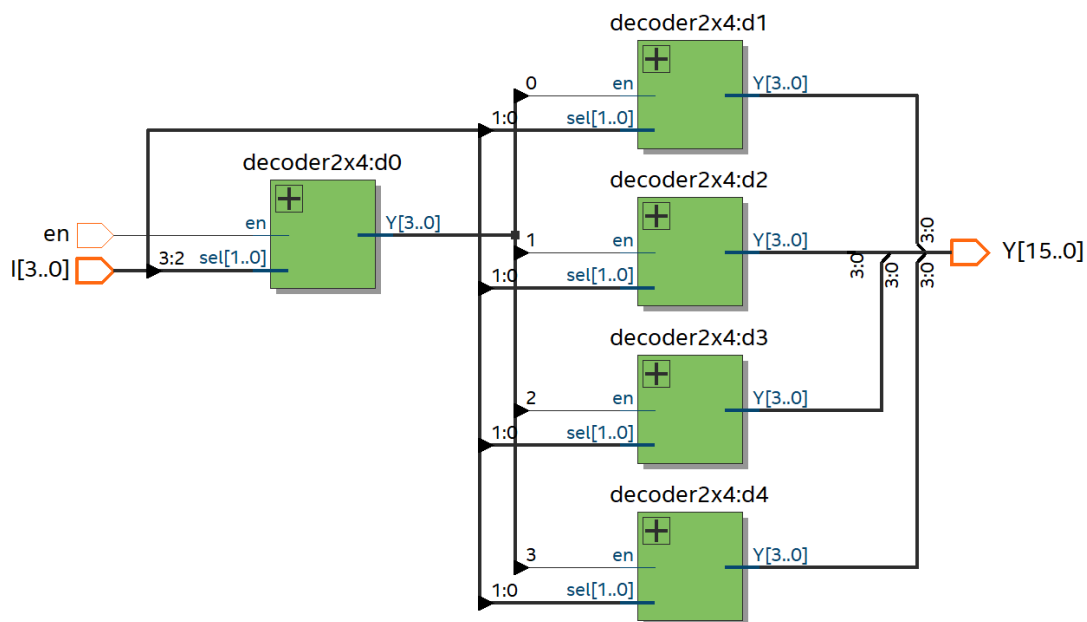
4:16 Decoder Tb

```
1 module tb_decoder4x16;
2     reg [3:0] I;
3     reg en;
4     wire [15:0]Y;
5     decoder4x16 dut(I,Y,en);
6     initial begin
7         $monitor("Time=%0t en=%b I=%b Y=%b", $time,en,I,Y);
8         en = 0; I = 4'b0000; #10;
9         en = 1;
10        I = 4'b0000; #10;
11        I = 4'b0101; #10;
12        I = 4'b1010; #10;
13        I = 4'b1111; #10;
14        $finish;
15    end
16 endmodule
```

Waveform



Schematic



EDA Tools Used

- IntelQuartusPrime
- ModelSim